

Compactness of the signed generator set

MathlibAnnex.PluckerBody.isCompact_generators

theorem

Passes compactness from contractions to their two signed minor images.

Statement

Let $n, N \in \mathbb{N}_0$ and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p : E \rightarrow \mathbb{R}$ be a continuous real seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in E).$$

The lower bound makes p a norm. Its closed unit ball and real reference volume are

$$K_p = \{x : p(x) \leq 1\}, \quad v_p = \lambda_n(K_p) \in (0, \infty).$$

Here and below a finite Lebesgue measure is read as a real number when used as a scalar.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B)$ for that vector scaled by v_p . Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_p(B) = v_p \Pi_n(B) \quad (I = (i_1 < \dots < i_n)).$$

A p -contraction is a linear map $B : E \rightarrow F$ satisfying $\|Bz\|_\infty \leq p(z)$ for every $z \in E$. Let $\mathcal{C}_p^N$ be their set, and put

$$\mathcal{G}_p^N = \{\mathcal{V}_p(B), -\mathcal{V}_p(B) : B \in \mathcal{C}_p^N\}.$$

Then $\mathcal{G}_p^N$ is compact.

Assumptions

The source and target are finite real coordinate spaces with sup norms. The seminorm p is continuous and has the stated positive lower and upper comparison constants. The measure is the original coordinate Lebesgue measure. The dimensions n, N may be zero. When $n = 0$, the unique empty minor has determinant 1; when $N < n$, the minor-index set is empty and its function space has one element. Neither case is excluded.

Conclusion

Compactness holds in the usual topology of the finite-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$.

Proof route

Bound the operators and then take their continuous signed images.

▼ Proof steps

1. **Make the contraction set compact.** Give the finite-dimensional operator space $\mathcal{L}(E, F)$ its induced operator norm

$$\|B\|_{\text{op}} = \sup_{\|x\|_{\infty} \leq 1} \|Bx\|_{\infty}.$$

Then

$$\mathcal{C}_p^N = \bigcap_{x \in E} \{B : \|Bx\|_{\infty} \leq p(x)\}, \quad \|B\|_{\text{op}} \leq b_p \quad (B \in \mathcal{C}_p^N).$$

For fixed x , evaluation $B \mapsto Bx$ is continuous, so every set in the intersection is closed. The norm bound follows from $\|Bx\|_{\infty} \leq p(x) \leq b_p \|x\|_{\infty}$. Thus the contraction set is closed and contained in the closed operator ball of radius b_p . Finite dimensionality makes it compact.

2. **Take two continuous images.** Put $Q = \mathcal{V}_p(\mathcal{C}_p^N)$. Then

$$\mathcal{G}_p^N = Q \cup (-Q).$$

This is the image-union identity. Matrix entries are continuous in the operator, determinants are finite polynomials, and the scalar v_p is fixed. These are continuity of maximal minors and continuity after volume scaling. The continuous-image theorem applied first to $\mathcal{C}_p^N$ and then to negation on Q makes Q and $-Q$ compact. Substitution into the displayed union gives compactness of $\mathcal{G}_p^N$.

Main citations

- Exact declaration and its source — `MathlibAnnex.PluckerBody.isCompact_generators`
- the image-union identity — `MathlibAnnex.PluckerBody.generators_eq_image_union`
- continuity of maximal minors — `MathlibAnnex.Matrix.continuous_maximalMinors`
- continuity after volume scaling —
`MathlibAnnex.Matrix.continuous_ballVolumeScaledMaximalMinors`
- Signed generator set — `MathlibAnnex.PluckerBody.generators`

Lean source signature (exact)

```
-- The generator set is compact at every pair of finite dimensions. -/  
theorem isCompact_generators {n N : ℕ} (M : EquivalentSeminorm (Fin n → ℝ)) :  
  IsCompact (generators M N)
```

There is no separate hypothesis after `M`. The conclusion begins after the final colon.

In the source	Mathematical meaning
<code>{n N : N}</code>	Arbitrary nonnegative integers n, N . Braces mean that Lean may infer these parameters; they are not extra assumptions.
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>M : EquivalentSeminorm (Fin n → ℝ)</code>	The continuous seminorm $p = M.p$, with $a_p = M.lower > 0$, $b_p = M.upper > 0$ and $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every $x \in E$.
<code>Matrix.MaximalMinorIndex n (Fin N)</code>	The index set $\mathcal{I}_{n,N}$: choose n distinct rows among the N rows and read them in increasing order. A function of this type into $\mathbb{R}$ is a minor-coordinate vector $z = (z_I)_{I \in \mathcal{I}_{n,N}}$.
<code>M.closedUnitBall; M.closedUnitBallVolume</code>	$K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p) > 0$, with the finite measure read as a real number.
<code>Matrix.ballVolumeScaledMaximalMinors M B</code>	$\mathcal{V}_p(B) = v_p \Pi_n(B)$, whose $I = (i_1 < \dots < i_n)$ coordinate is $v_p \det ((Be_j)_{i_a})_{a,j=1}^n$.

In the source	Mathematical meaning
<code>generators M N</code>	The set $\mathcal{G}_p^N = \{\pm v_p \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z \in E, \ Bz\ _\infty \leq p(z)\}$. Its elements are minor-coordinate vectors, not the maps B .
<code>IsCompact (generators M N)</code>	The set $\mathcal{G}_p^N$ is compact in the finite-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ with its usual topology.

Exact definitions used in this reading: Signed generator set.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–14:

```
noncomputable section

namespace MathlibAnnex.PluckerBody
```

▼ Exact content identity

Declaration: `MathlibAnnex.PluckerBody.isCompact_generators`

Accepted content SHA-256: f6651ef68f8f256e9e59b470b843a0e15d459ca92a14da43067073b4f682cef4

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Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73